

PRACTICAL : WEEK-5

Task - 1:

Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency.

CODE:

```
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <time.h>

int main()
{
    int fd[2];
    pid_t pid;
    int data;

    struct timespec start, end;

    pipe(fd);

    pid = fork();

    if (pid == 0)
    {
        // Child - Consumer

        close(fd[1]);

        printf("Child (Consumer): Receiving data...\n");

        while (read(fd[0], &data, sizeof(data)) > 0)
        {
            printf("Consumed: %d\n", data);
        }

        close(fd[0]);

        return 0;
    }
    else
    {
        // Parent - Producer

        close(fd[0]);

        clock_gettime(CLOCK_MONOTONIC, &start);

        printf("Parent (Producer): Sending data...\n");

        for (int i = 1; i <= 100; i++)
        {
            write(fd[1], &i, sizeof(i));
        }

        close(fd[1]);

        wait(NULL);
        clock_gettime(CLOCK_MONOTONIC, &end);

        double time_taken =
            (end.tv_sec - start.tv_sec) +
            (end.tv_nsec - start.tv_nsec) / 1e9;

        printf("\nCommunication completed.\n");
        printf("Communication time: %f seconds\n", time_taken);

        return 0;
    }
}
```

OUTPUT :

```
ananya@Ananya:/mnt/d/OSSP_Programs/Practical/Week5$ nano producer_consumer.c
ananya@Ananya:/mnt/d/OSSP_Programs/Practical/Week5$ gcc producer_consumer.c -o producer_consumer
ananya@Ananya:/mnt/d/OSSP_Programs/Practical/Week5$ ./producer_consumer
Parent (Producer): Sending data...
Child (Consumer): Receiving data...
Consumed: 1
Consumed: 2
Consumed: 3
Consumed: 4
Consumed: 5
Consumed: 6
Consumed: 7
Consumed: 8
Consumed: 9
Consumed: 10
Consumed: 11
Consumed: 12
Consumed: 13
Consumed: 14
Consumed: 15
Consumed: 16
Consumed: 17
Consumed: 18
Consumed: 19
Consumed: 20
Consumed: 21

Consumed: 94
Consumed: 95
Consumed: 96
Consumed: 97
Consumed: 98
Consumed: 99
Consumed: 100

Communication completed.
Communication time: 0.000838 seconds
```

OBSERVATION:

The parent process successfully acted as the producer and generated data, which was sent to the child process through an anonymous pipe. The child process acted as the consumer and successfully received and displayed the data using the read end of the pipe. The communication between the parent and child processes was successfully established using `pipe()` and `fork()`. The time taken for the producer-consumer communication was measured using `clock_gettime()`, allowing the communication efficiency to be observed. The experiment demonstrates that anonymous pipes can be effectively used for communication between related processes.

Task - 2:

Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

CODE:

```
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int fd[2];
    pid_t pid1, pid2;

    pipe(fd);

    // First child - executes ls -l
    pid1 = fork();

    if (pid1 == 0)
    {
        close(fd[0]);

        dup2(fd[1], STDOUT_FILENO);
        close(fd[1]);

        execlp("ls", "ls", "-l", NULL);

        return 0;
    }

    // Second child - executes grep ".c"
    pid2 = fork();

    if (pid2 == 0)
    {
        close(fd[1]);

        dup2(fd[0], STDIN_FILENO);
        close(fd[0]);

        execlp("grep", "grep", ".c", NULL);

        return 0;
    }

    // Parent
    close(fd[0]);
    close(fd[1]);

    waitpid(pid1, NULL, 0);
    waitpid(pid2, NULL, 0);

    return 0;
}
```

OUTPUT:

```
ananya@Ananya:/mnt/d/OSSP_Programs/Practical/Week5$ pwd
/mnt/d/OSSP_Programs/Practical/Week5
ananya@Ananya:/mnt/d/OSSP_Programs/Practical/Week5$ nano pipeline.c
ananya@Ananya:/mnt/d/OSSP_Programs/Practical/Week5$ gcc pipeline.c -o pipeline
ananya@Ananya:/mnt/d/OSSP_Programs/Practical/Week5$ ./pipeline
-rwxrwxrwx 1 ananya ananya  718 Sep  9 03:04 pipeline.c
-rwxrwxrwx 1 ananya ananya 16352 Sep  9 03:01 producer_consumer
-rwxrwxrwx 1 ananya ananya  1128 Sep  9 03:01 producer_consumer.c
ananya@Ananya:/mnt/d/OSSP_Programs/Practical/Week5$ |
```

OBSERVATION:

The program successfully implemented the equivalent of the shell command `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls. The first child process executed `ls -l` and redirected its output to the pipe, while the second child process received the pipe output as its input and executed `grep ".c"`. The required `.c` files were successfully displayed as the output. The parent process used `waitpid()` to synchronize with both child processes before terminating. Thus, inter-process communication and command piping were successfully demonstrated.